

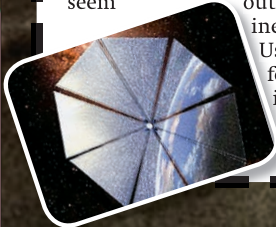


## geek life

A missile like the Jericho missile would be too impractical for real-world military use.

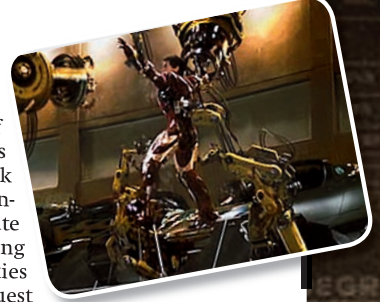
### Personal Jets and Repulsor technology

Iron Man's boots have jets that enable him to fly. There is one glaring omission here – where does the fuel come from? Any kind of jet requires fuel, and Iron Man is not bundled with a fuel tank for transatlantic travel. There are jetpacks that exist (though none that are shoe-based), and some of them are even available to laymen. The problem here is that these jetpacks can carry only enough fuel for very short flights – usually less than a minute. Jetpacks are also used by astronauts in space, but the fuel requirements of zero gravity is far less than the fuel requirements in the atmosphere. Iron Man's palms use repulsor technology to allow Iron Man to balance on his jets and navigate while in flight. This is almost certainly some kind of particle generator with a high density of small particles. From the way it shines, it is almost certainly light-based. While this may seem outrageous, it is actually more genuine scientific concept than the jets. Using a high concentration of light for movement is not unheard of, and is the technology behind solar sails, a proposed method of low-fuel long distance space travel.



### Jarvis

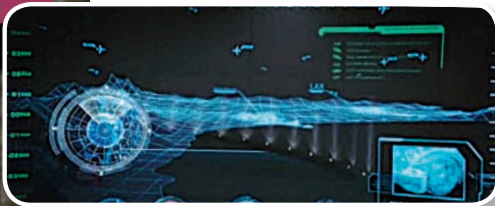
Jarvis is the AI in Tony Stark's house and controls all the robots, interfaces and handles a number of operations. The capabilities of Jarvis are vast. He can hack into commercial and government air bases, and calculate the trajectories of all the flying objects in the nearby skies (although, Jarvis does request some time to do this). Jarvis can also translate languages on the fly, interpret actions or gestures, process complex instructions, and practically read minds of the humans around it. There are a number of independent efforts in developing AI. In fact, a fair penetration of the implementations, from air bags in cars to temperature controllers in refrigerators. Most of the AI however, are very complex algorithms based on continuous or even real-time feedback. However, no one has come close to the complex AI depicted in Iron Man. The kind of processing power involved simply does not exist, nor any real framework for the programming. If you are interested, you can head over to [www.personalityforge.com](http://www.personalityforge.com) and create your own AI to understand the processes involved.



### Interfaces

The 3D interface used by Tony Stark to design his Iron Man suit was a step up from the tactile interface shown in Minority Report. While hologram projection may not exist, a tactile 3D interface on a 2D touchscreen has been around for some time now. CityWall is one such multi-touch interface, with videos all over the internet. In a particular scene, Stark drags a window from one monitor onto another using gestures. This is similar to dragging windows between work spaces on some existing operating systems. Gesture-based computing is not new

at all, and the US military is at the forefront of research. The many semi-transparent touch displays may not be far away either – MERL and Microsoft Research are working on LucidTouch technology that allows navigation of a device through its back instead of the front, which makes holding the device easier. The HUD (heads up display) in Iron Man's helmet has been around for a long time, particularly used by fighter pilots who can keep their eyes on the skies instead of the instrument panel on the planes for vital information. There are a whole bunch of commercial motorcycle helmets with an HUD you can purchase, including the Vizalert, which shows any speedguns in the peripheral vision.



### Get Real

In real life, which of the two heroes are more likely to exist? Batman's tumbler is actually a very impractical vehicle, mostly because it would be flimsier than it is shown to be. Iron Man's armour, also simply cannot exist because "repulsor" is not even a word, and the fuel for the jets would be impractical. Batman's kevlar suit, while being able to stop bullets, cannot stop him from being injured after long falls, which happens at least twice in the movie. Iron Man's ammunition hidden away in his armour is also an impractical thing. Tiny little missiles blowing up tanks? No way. In fact, none of the technology depicted in the movie would have been practical in real life... so to find an answer, we have to explore the roots of the superheroes; stripping essentials. Batman is just a man in a cape, with a love for gadgets... smoke bombs, some rope, and a jet powered car. Iron Man however, had jets on his boots and an iron exoskeleton forever. Tony Stark is a borderline renegade genius, but Batman was a man on a mission, who trained his body for everything he does. This round clearly goes to Batman.

